



COURSE CODE/TITLE: CPE212 | Automating Server Management

SECTION: CPE31S2

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ACTIVITY NO. 1.1 Hands-on Activity - Creating Virtual Machines

1. PROCEDURE OUTPUT

What procedures did you perform?

TASK 1:

```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ sudo nano /etc/hostname  
ubuntu@ubuntu:~$ sudo nano /etc/hosts  
ubuntu@ubuntu:~$
```

```
GNU nano 7.2 /etc/hostname  
server1
```

```
GNU nano 7.2 /etc/hosts
127.0.0.1 server1
127.0.1.1 ubuntu

# The following lines are desirable for IPv6 capable hosts
::1    ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```

```
ubuntu@ubuntu: ~
GNU nano 7.2 /etc/hostname
server2
```

```
ubuntu@ubuntu: ~
GNU nano 7.2 /etc/hosts
127.0.0.1 server2
127.0.1.1 ubuntu

# The following lines are desirable for IPv6 capable hosts
::1    ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```

```
ubuntu@ubuntu: ~
GNU nano 7.2 /etc/hosts *
127.0.0.1 localmachine
127.0.1.1 ubuntu

# The following lines are desirable for IPv6 capable hosts
::1    ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```

```
ubuntu@ubuntu:~$ sudo nano /etc/hosts
ubuntu@ubuntu:~$ sudo apt update
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
Release
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1079 k
B]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [324 k
B]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [515 kB
]

ubuntu@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm19 libwoff1
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libllvm20 linux-headers-6.17.0-35-generic linux-hwe-6.17-headers-6.17.0-35
  linux-hwe-6.17-tools-6.17.0-35 linux-image-6.17.0-35-generic
  linux-modules-6.17.0-35-generic linux-modules-extra-6.17.0-35-generic
  linux-tools-6.17.0-35-generic
The following upgrades have been deferred due to phasing:
  fwupd gir1.2-mutter-14 libmutter-14-0 mutter-common mutter-common-bin
The following packages have been kept back:
  libgl1-amber-dri
The following packages will be upgraded:
  alsa-ucm-conf amd64-microcode apparmor apport apport-core-dump-handler
  apport-gtk apt apt-utils avahi-daemon base-files bind9-dnsutils bind9-host
  bind9-libs bluez bluez-cups bluez-obexd bsdxtrutils bsdxtrutils
  ca-certificates cifs-utils cloud-init coreutils cpp-13
```

Server 2:

```
ubuntu@ubuntu:~$ sudo nano /etc/hostname
ubuntu@ubuntu:~$ sudo apt update
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
Release
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]

638 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libgl1-amd-gpu-dri libglapi-mesa libllvm19 libwoff1
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libllvm20 linux-headers-6.17.0-35-generic linux-hwe-6.17-headers-6.17.0-35
  linux-hwe-6.17-tools-6.17.0-35 linux-image-6.17.0-35-generic
```

Localmachine:

```
ubuntu@ubuntu:~$ sudo nano /etc/hosts
ubuntu@ubuntu:~$ sudo apt update
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
Release
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
0% [7 InRelease 4072 B/126 kB 3%]
```

```
python3-urllib3 python3.12 python3.12-minimal remmina remmina-common
remmina-plugin-rdp remmina-plugin-secret remmina-plugin-vnc rkill rsync
rsyslog samba/libs sed simple-scan snapd software-properties-common
software-properties-gtk sssd sssd-ad sssd-ad-common sssd-common sssd-ipa
sssd-krb5 sssd-krb5-common sssd-ldap sssd-proxy sudo systemd systemd-dev
systemd-hwe-hwdb systemd-oemd systemd-resolved systemd-sysv
systemd-timesyncd tar tcpdump tecla telnet thermald tracker-extract
tracker-miner-fs transmission-common transmission-gtk tzdata
ubuntu-drivers-common ubuntu-pro-client ubuntu-pro-client-l10n
ubuntu-release-upgrader-core ubuntu-release-upgrader-gtk ubuntu-report
ubuntu-settings udev udisks2 uno-libs-private update-manager
update-manager-core update-notifier update-notifier-common ure usbmuxd
util-linux uuid-runtime vim-common vim-tiny whoopsie wireless-regdb
wireplumber wpasupplicant xdg-dbus-proxy xdg-desktop-portal xserver-common
xserver-xephyr xserver-xorg-core xserver-xorg-input-libinput
xserver-xorg-legacy xserver-xorg-video-amdgpu xserver-xorg-video-nouveau
xserver-xorg-video-vesa xwayland xxd xz-utils yelp yelp-xsl zfs-zed
zfsutils-linux
631 upgraded, 9 newly installed, 0 to remove and 7 not upgraded.
388 standard LTS security updates
Need to get 1460 MB of archives.
After this operation, 690 MB of additional disk space will be used.
N: Some packages may have been kept back due to phasing.
Do you want to continue? [Y/n]
```

```
ubuntu@ubuntu:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libncurses6 libncursesw6 libtinfo6 ncurses-base ncurses-bin
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  libncurses6 libncursesw6 libtinfo6 ncurses-base ncurses-bin
  openssh-client
6 upgraded, 4 newly installed, 0 to remove and 632 not upgraded.
Need to get 832 kB/2321 kB of archives.
After this operation, 6748 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

```

ubuntu@ubuntu:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libncurses6 libncursesw6 libtinfo6 ncurses-base ncurses-bin
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  libncurses6 libncursesw6 libtinfo6 ncurses-base ncurses-bin
  openssh-client
6 upgraded, 4 newly installed, 0 to remove and 632 not upgraded.
Need to get 832 kB/2321 kB of archives.
After this operation, 6748 kB of additional disk space will be used
Do you want to continue? [Y/n]

```

```

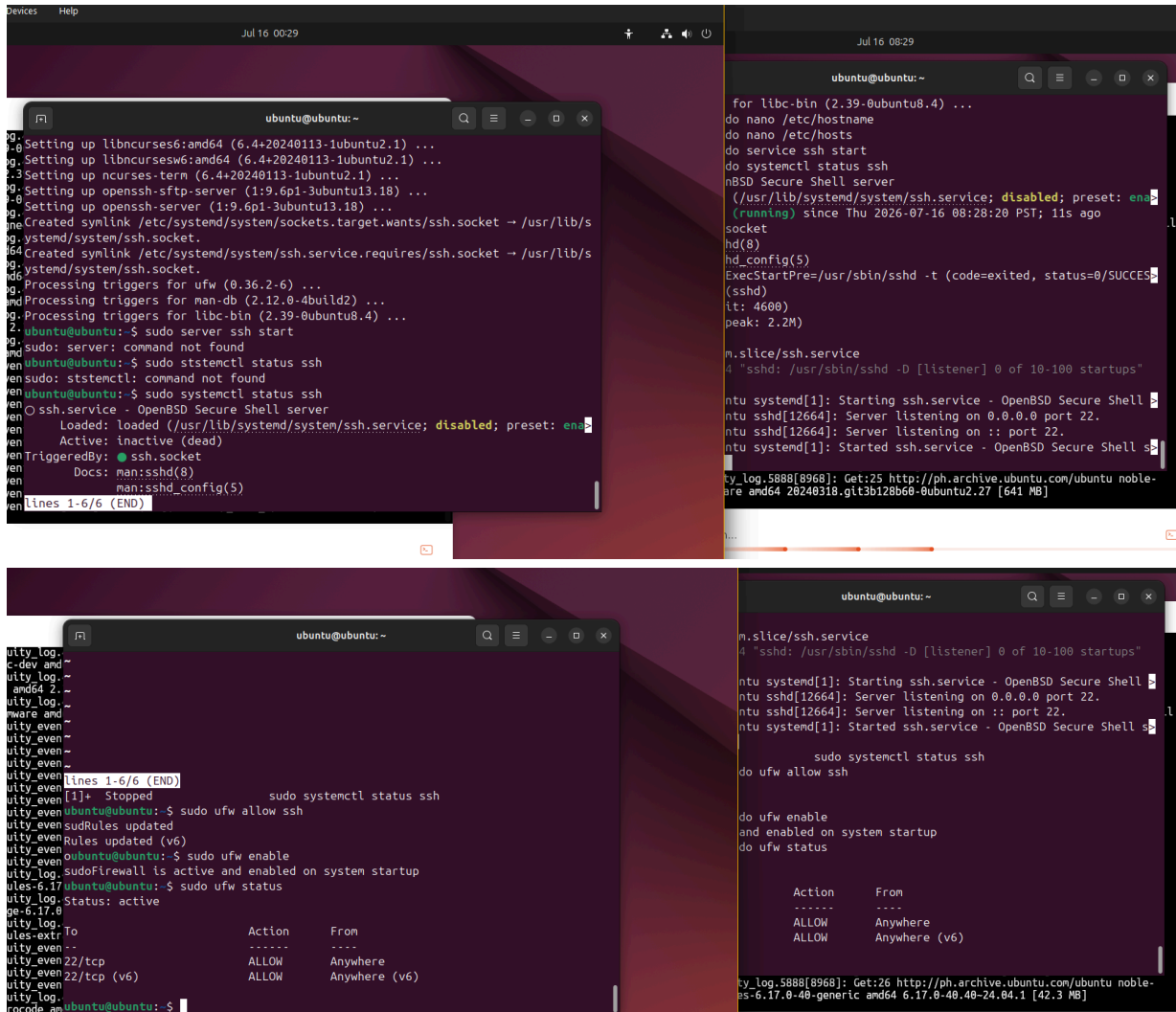
ubuntu@ubuntu:~$ sudo apt install openssh-server
Setting up libtinfo6:amd64 (6.4+20240113-1ubuntu2.1) ...
Setting up previously unselected package ncurses-term.
(Reading database ... 212060 files and directories currently installed.)
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu2.1_all.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu2.1) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2.24.04.1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up openssh-client (1:9.6p1-3ubuntu13.18) ...
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up libncurses6:amd64 (6.4+20240113-1ubuntu2.1) ...
Setting up libncursesw6:amd64 (6.4+20240113-1ubuntu2.1) ...
Setting up ncurses-term (6.4+20240113-1ubuntu2.1) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.18) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.18) ...
Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket → /usr/lib/s
systemd/system/ssh.socket.
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/s
systemd/system/ssh.socket.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Progress: [ 98%] [#####]

```

```

ubuntu@ubuntu:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libncurses6 libncursesw6 libtinfo6 ncurses-base ncurses-bin
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  libncurses6 libncursesw6 libtinfo6 ncurses-base ncurses-bin
  openssh-client
6 upgraded, 4 newly installed, 0 to remove and 632 not upgraded.
Need to get 832 kB/2321 kB of archives.
After this operation, 6748 kB of additional disk space will be used
Do you want to continue? [Y/n]

```



```

Setting up libncurses6:amd64 (6.4+20240113-1ubuntu2.1) ...
Setting up libncursesw6:amd64 (6.4+20240113-1ubuntu2.1) ...
Setting up ncurses-term (6.4+20240113-1ubuntu2.1) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.18) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.18) ...
Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket → /usr/lib/s
systemd/system/ssh.socket.
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/s
systemd/system/ssh.socket.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
ubuntu@ubuntu:~$ sudo server ssh start
sudo: server: command not found
ubuntu@ubuntu:~$ sudo systemctl status ssh
sudo: systemctl: command not found
ubuntu@ubuntu:~$ sudo systemctl status ssh
ssh.service - OpenBSD Secure Shell server
Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena
Active: inactive (dead)
TriggeredBy: ● ssh.socket
Docs: man:sshd(8)
man:sshd_config(5)
lines 1-6/6 (END)

m.slice/ssh.service
4 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

ntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell
ntu sshd[12664]: Server listening on 0.0.0.0 port 22.
ntu sshd[12664]: Server listening on :: port 22.
ntu systemd[1]: Started ssh.service - OpenBSD Secure Shell s

ty_log.5888[8968]: Get:25 http://ph.archive.ubuntu.com/ubuntu noble-
as-6.17.0-generic amd64 6.17.0-40.40-24.04.1 [42.3 MB]

m.slice/ssh.service
4 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

ntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell
ntu sshd[12664]: Server listening on 0.0.0.0 port 22.
ntu sshd[12664]: Server listening on :: port 22.
ntu systemd[1]: Started ssh.service - OpenBSD Secure Shell s

do systemctl status ssh
do ufw allow ssh

do ufw enable
and enabled on system startup
do ufw status

Action From
-----
ALLOW Anywhere
ALLOW Anywhere (v6)

ty_log.5888[8968]: Get:26 http://ph.archive.ubuntu.com/ubuntu noble-
as-6.17.0-generic amd64 6.17.0-40.40-24.04.1 [42.3 MB]
  
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Reflections:

Answer the following:

- How are we able to use the hostname instead of IP address in SSH commands?
 - We are able to use the hostname instead of the IP address in SSH because the OS relies on name resolution systems to translate human readable names into the numerical IP address for it to route.
- How secured is SSH?
 - It's secure and serves as the standard for managing remote servers, but as usual it depends on how you configure it and how you manage the access of the credentials.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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